Chapter

2

Chapter Learning Outcomes

CLO

2

- Determine if a given sentence is a proposition or not.
- Complete and use truth tables for expressions involving the following logical connectives: negation, conjunction, disjunction, conditional, and biconditional.
- Write the converse, contrapositive and inverse for a given sentence.
- Translate an English sentence to PL and vice-versa.
- Define and determine validity, inconsistency, and satisfiability for a PL expression.
- Prove if two propositions are logically equivalent or not.
- Understand the concept of Entailment.
- List all models of a knowledge base.
- Convert a proposition to CNF.
- Apply various methods for inferencing (enumeration, inference rules, resolution refutation).

Propositional Logic

The field of mathematics known as symbolic logic, in which symbols are used to replace ordinary language.

Proposition

A **proposition**, or *statement*, is a declarative sentence that can be classified as either true or false but not both.

Propositional Logic

Proposition

A **proposition**, or *statement*, is a declarative sentence that can be classified as either true or false but not both.

Example : All of the following are **propositions**:

a) This course is Artificial Intelligence.
b) Today is Friday.
c) Amman is the capital of Jordan.
d) $1 + 0 = 1$
e) $0 + 0 = 2$

Propositional Logic

Proposition

A **proposition**, or *statement*, is a declarative sentence that can be classified as either true or false but not both.

Example : All of the following are **not propositions**:

a) Sit down!
b) What is the time now?
c) $x + 1 = 2$
d) $x + y = z$
e) Either 7 is an odd number or it is even

Propositional Logic

The words that are used to combine propositions are called logical connectives. The basic connectives that we will consider are given in Table 1 along with their symbols.

TABLE 1

Name	Logical Connective	Symbol
Conjunction	and	$\wedge$
Disjunction	or	$\vee$
Negation	not	$\sim$
Conditional	if . . . then	$\rightarrow$
Biconditional	if and only if	$\leftrightarrow$

Conjunction

Conjunction

A **conjunction** is a statement of the form "p and q" and is represented symbolically by

$$p \wedge q$$

The conjunction $p \wedge q$ is true if *both* p and q are true; it is false otherwise.

p	q	p ∧ q
T	T	T
T	F	F
F	T	F
F	F	F

How to Write Logical Connectives Using Keyboard

and — /\ (slash then backslash)

Disjunction

Disjunction

A **disjunction** is a proposition of the form "p or q" and is represented symbolically by

$$p \vee q$$

The disjunction $p \vee q$ is false if *both* statements p and q are false; it is true in all other cases.

p	q	p Vq
T	T	T
T	F	T
F	T	T
F	F	F

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or $\vee$ (backslash then slash)

Negation

Negation

A **negation** is a proposition of the form "not p" and is represented symbolically by

$$\sim p$$

The proposition $\sim p$ is true if p is false and vice versa.

p	¬p
T	F
F	T

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Negation

~ (tilde symbol)

Conditional Statement

Conditional Statement

A **conditional statement** is a proposition of the form "if p, then q" and is represented symbolically by

$$p \rightarrow q$$

The connective "if . . . then" is called the **conditional connective**; the proposition p is called the **hypothesis**; and the proposition q is called the **conclusion**. A conditional statement is false if the hypothesis is true and the conclusion is false; it is true in all other cases.

p	q	p →q
T	T	T
T	F	F
F	T	T
F	F	T

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Implies -> (dash then greater than symbols)

Different Forms for Implications

"if H then C" "H → C" is equivalent to :

1- if H, C

2- H implies C

3- H only if C

4- C if H

5- C when H

6- C whenever H

7- C unless ~H

8- H is sufficient for C

9- C is necessary for H

10- a necessary condition for H is C

11- a sufficient condition C is H

12- C follows from H

Converse, Contrapositive and Inverse

We refer to statements that are variations of the conditional H → C as logical variants. We define three logical variants of the conditional: H → C as follows:

• Converse	if C, then H	C → H
• Contrapositive	if not C, then not H	¬ C → ¬ H
• Inverse	if not H, then not C	¬ H → ¬ C

Converse, Contrapositive and Inverse

$$H \rightarrow C$$

Inverse: $\neg H \rightarrow \neg C$

Converse: $C \rightarrow H$

Contrapositive: $\neg C \rightarrow \neg H$

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Converse, Contrapositive and Inverse

EXAMPLE

Write the converse, the contrapositive, and the inverse for the following conditional statement:

$$p \rightarrow \neg q$$

Inverse

Converse

Contrapositive

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Converse, Contrapositive and Inverse

EXAMPLE

Write the converse, the contrapositive, and the inverse for the following conditional statement:

If n^2 is an odd number, then n is an odd number

Inverse

Converse

Contrapositive

Converse, Contrapositive and Inverse

EXAMPLE

Write the converse, the contrapositive, and the inverse for the following conditional statement:

If it rains, then they cancel the school

$H \rightarrow C$

Inverse

Converse

Contrapositive

Biconditional Statement

Biconditional Propositions

Statements of the form "p if and only if q" are called **biconditional propositions** and are represented symbolically by

$$p \longleftrightarrow q$$

The connective "if and only if" is called the *biconditional connective*. The biconditional $p \longleftrightarrow q$ is true whenever p and q are *both true* or *both false*.

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Same as **Exclusive-NOR**

$\overline{A \oplus B}$

How to Write Logical Connectives Using Keyboard

iff

<-> (Less than then dash then greater than)

Different Forms for Biconditional

"**p if and only if q**" "**P ↔ q**" is equivalent to :

1- *p* **iff** *q*

2- *p* **is necessary and sufficient for** *q*

3- if p then q , and conversely

Order of Precedence of Logical Operators

$$(\,),\quad \neg,\quad \wedge,\quad \vee,\quad \rightarrow,\quad \leftrightarrow,\quad \equiv$$

$$\underbrace{\underbrace{p \vee q}_{2} \rightarrow \underbrace{\neg r}_{1}}_{3}$$

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Truth Tables For Compound Propositions

Example: Construct a truth table for $p \vee q \rightarrow \neg r$

1- How many atomic sentences are there in the expression?

1- How many compound sentences are there in the expression?

Truth Tables For Compound Propositions

Example: Construct a truth table for $p \vee q \rightarrow \neg r$

- No. of rows in the table=
- No. of columns in the table=

p	q	r	$\neg r$	$p \vee q$	$p \vee q \rightarrow \neg r$
T	T	T			
T	T	F			
T	F	T			
T	F	F			
F	T	T			
F	T	F			
F	F	T			
F	F	F			

Truth Tables For Compound Propositions

Exercise: Construct a truth table for the following sentences. In each question determine the number of atomic and compound sentences.

1- $\sim(p \rightarrow q) \wedge p$

2- $(p \rightarrow q) \leftrightarrow (\sim p \vee q)$

3- $[(p \rightarrow q) \vee (q \rightarrow r)] \rightarrow (p \rightarrow r)$

Translating English Sentences to PL

Amman is the capital of Jordan and it is the largest city.

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Translating English Sentences to PL

Ali likes fishing and reading.

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Translating English Sentences into PL

Jones is a fool **or** a liar

Jones is **neither** a fool **nor** a liar

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Translating English Sentences into PL

Ali can not play football today.

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Translating English Sentences into PL

If it rains, then the baseball game will be postponed.

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Translating English Sentences to PL

Ali is going to the university if and only if Ahmad is going to the university.

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Translating English Sentences to PL

Let p and q be the propositions:

- p : The stock market goes up.
- q : Interest rates decrease.

Represent the following statements using PL:

1- If interest rates decrease, then the stock market goes up.

2- If interest rates do not decrease, then the stock market does not go up.

Translating English Sentences to PL

Let p and q be the propositions:

- p : The stock market goes up.
- q : Interest rates decrease.

Represent the following statements using PL:

3- The stock market will go up if and only if interest rates decrease.

4- Decreasing interest rates is a sufficient condition for the stock market to go up.

Translating English Sentences to PL

If a number is divisible by 10, then it is divisible by 2

$p \rightarrow q$	
where:	
p	A number is divisible by 10
q	A number is divisible by 2

Translating English Sentences to PL

x is a sufficient condition for y.

x is a necessary condition for y.

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Translating English Sentences to PL

x is a necessary and a sufficient condition for y.

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Translating English Sentences into PL

It is raining, when Ali is come.

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Translating English Sentences into PL

You cannot ride the roller coaster if you are under 4 feet tall unless you are older than 16 years old.

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Quiz

Translating the following sentence into PL:

If it is sunny today, then the sun shines on the screen. If the sun shines on the screen, the blinds are brought down. The blinds are not down.

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Quiz

Translating the following sentence into PL:

If it is humid, then it is hot. If it is hot and humid, then it is raining.

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Quiz

Translating the following sentence into PL:

You can access the Internet from campus only if you are a computer science major or you are not a freshman.

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Quiz

Lets consider a propositional logic where :

A =”Ali comes to the party”.

B =”Basem comes to the party”.

C =”Kamal comes to the party”.

D =”Dalal comes to the party”.

Translate the following English sentences to propositions logic.

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Quiz … cont.

1. If Dalal comes to the party then Basem and Kamal come too.

2. Kamal comes to the party only if Ali and Basem do not come.

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Quiz … cont.

3. If Dalal comes to the party, then, if Kamal doesn't come then Ali comes

4. Kamal comes to the party provided that Dalal doesn't come, but, if Dalal comes, then Basem doesn't come.

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Quiz … cont.

5. A necessary condition for Ali coming to the party, is that, if Basem and Kamal aren't coming, Dalal comes.

6. Ali, Basem and Kamal come to the party if and only if Dalal doesn’t come, but, if neither Ali nor Basem come, then Dalal comes only if Kamal comes.

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Tautologies, Contradictions, and Contingencies

- A **tautology (valid)** is a proposition which is **always true**.
  Example: $p \vee \neg p$

- A **contradiction (Unsatisfiable)(Inconsistent)** is a proposition which is **always false**.
  Example: $p \wedge \neg p$

- A **contingency (Satisfiable)** is a proposition which **is neither a tautology nor a contradiction,** such as p.

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Tautologies, Contradictions, and Contingencies

1- Prove that the following proposition is tautology

$$p \vee \neg p$$

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2- Prove that the following proposition is contradiction

$$p \wedge \neg p$$

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3- Prove that the following proposition is contingency.

$$\neg p \vee q$$

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Tautologies, Contradictions, and Contingencies

3- Check if the following proposition is tautology, contingency or contradictory.

$$(p \rightarrow r) \vee (q \rightarrow r)$$

p	q	r	$p \rightarrow r$	$q \rightarrow r$	$(p \rightarrow r) \vee (q \rightarrow r)$
F	F	F			
F	F	T			
F	T	F			
F	T	T			
T	F	F			
T	F	T			
T	T	F			
T	T	T			

Logically Equivalent

Two compound propositions p and q are **equivalent** if and only if **the columns in a truth table giving their truth values agree**.

Prove that the following propositions are **equivalent**

$$\neg p \vee q \equiv p \rightarrow q$$

p	q	$\neg p$	$\neg p \vee q$	$p \rightarrow q$
T	T	F		
T	F	F		
F	T	T		
F	F	T		

Quiz

Prove that the following propositions are equivalent

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

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Quiz

Prove that the following statement is **tautology**.

$$P \wedge \neg P \rightarrow Q$$

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Quiz

Prove that the following statement is **contradiction**.

$$\neg(\neg(P \vee Q) \leftrightarrow (\neg P) \wedge (\neg Q))$$

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Entailment

Entailment means that one thing follows from another.

Knowledge Base (KB): a collection of sentences.

KB entails S, written **KB |= S**, means that whenever KB is True, so is S.

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Entailment

Assume you have a Knowledge Base (KB), and a query sentence (S). How to check if KB |= S.

- **If KB → S is valid / tautology then KB |= S.**
- **If KB ∧ ¬S is contradiction then KB |= S**

Entailment Example 1

EXAMPLE

- Assume KB has the following sentences:

  1- $A \vee B$ 2- $\neg C \vee A$
- Assume the query (S) is: $A \vee B \vee C$
- Prove that KB |= S.

See Next Slide for Solution

Entailment Example

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Entailment Example 2

EXAMPLE

- Assume KB has the following sentences:
  1- $A \vee B$ 2- $\neg C \vee A$
- Assume the query (S) is: $A \wedge C$
- Prove that KB |= S.

- If KB $\rightarrow$ S is valid / tautology then KB |= S.
- If KB $\wedge \neg$S is contradiction then KB |= S

See Next Slide for Solution

Entailment Example

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A Model

Definition (model)

A **model** of a knowledge base KB is an interpretation in which every clause in KB is true.

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A Model

EXAMPLE

- Assume KB has the following sentences:

  1- $A \vee B$

List all the models of KB

A	B	KB: $A \vee B$	Is Model?
T	T		
T	F		
F	T		
F	F		

Quiz

- Assume KB has the following sentences:

  1- $q \rightarrow p$ 2- q 3- $s \rightarrow r$
- List all the models of KB

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Inference

An **inference** algorithm is a procedure for deriving a new sentence from the KB.

Basic problem:

- We have a KB
- We have a sentence S (the “query”)
- We want to check KB |= S

Inference Methods

- Inference by **Enumeration (Truth Table)**.
- Inference by **Inference Rules**.
- Inference by **Resolution Refutation.**

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Method 1: Inference By Enumeration

- Using inference by enumeration to build a complete truth table in order to determine if a sentence is entailed by KB.
- Also called Model Checking or Truth Table Enumeration.
- Evaluate all possible settings of the symbols.
- This method is very slow: takes exponential time.

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Method 1: Inference By Enumeration

EXAMPLE

- Assume KB has the following sentences:

  1- $p \rightarrow q$ 2- $q \rightarrow r$
- $S = p \rightarrow r$
- Prove that KB |= S using the enumeration method.

- If KB $\rightarrow$ S is valid / tautology then KB |= S.

See Next Slide for Solution

Method 1: Inference By Enumeration

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Quiz

- Assume KB has the following sentences:

  1- $B \wedge C \rightarrow A$

  2- B

  3- $D \wedge E \rightarrow C$

  4- $E \vee F$

  5- $D \wedge \neg F$

- Prove that $KB \models A$ using the enumeration method.

Quiz

- Assume KB has the following sentences:
  1- $P \rightarrow R$
  2- $Q \rightarrow R$
  3- $P \vee Q$

- Prove that KB $\models$ R using the enumeration method.

Quiz

- Assume KB has the following sentences:

  1- $(P \rightarrow Q) \rightarrow Q$

  2- $(P \rightarrow P) \rightarrow R$

  3- $(R \rightarrow S) \rightarrow \neg(S \rightarrow Q)$

- Prove that KB $\models$ R using the enumeration method.

Method 2: Inference By Inference Rules

Logical Equivalences

1-	Identity laws	$p \wedge T \equiv p$ $p \vee F \equiv p$
2-	Domination laws	$p \wedge F \equiv F$ $p \vee T \equiv T$
3-	Idempotent laws	$p \wedge p \equiv p$ $p \vee p \equiv p$
4-	Commutative laws	$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$
5-	Associative laws	$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
6-	Distributive laws	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
7-	De Morgan's laws	$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$

Method 2: Inference By Inference Rules

Logical Equivalences

8-	Absorption laws	$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$
9-	Negation laws	$p \wedge \neg p \equiv F$ $p \vee \neg p \equiv T$
10-	Implication Elimination	$p \rightarrow q \equiv \neg p \vee q$
11-	Contraposition law	$p \rightarrow q \equiv \neg q \rightarrow \neg p$
12-	Biconditional Elimination	$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$
13-	Modus Ponens	$\dfrac{p \rightarrow q, p}{q}$

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Method 2: Inference By Inference Rules

EXAMPLE

- Assume KB has the following sentences:
  1- $(P \wedge \neg Q) \vee Q$
- Prove that KB $\models P \vee Q$ using the inference rules.

Sentence	Reason

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Method 2: Inference By Inference Rules

EXAMPLE

- Prove : $(P \wedge Q) \rightarrow P$ is tautology using inference rules.

Sentence	Reason

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Quiz

- Assume KB has the following sentences:
  1. $P \wedge Q \rightarrow R$
  2. $S \wedge E \rightarrow Q$
  3. S
  4. E
  5. P

- Prove that KB |= R using the inference rules..

Quiz - Solution

Sentence	Reason

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Method 3: Inference By Resolution Refutation

- Take any two clauses where one contains some symbol, and the other contains its complement (negative).
- Merge (resolve) them, throw away the symbol and its complement.
- If two clauses resolve and there's no symbol left, you have derived the empty clause (False), so KB |= S.
- If no new clauses can be added, KB $\not\models$ entail S.

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Method 3: Inference By Resolution Refutation

Resolution Refutation Rule

$$\frac{A \vee B\ ,\ \neg B \vee C}{A \vee C}$$

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Method 3: Inference By Resolution Refutation

EXAMPLE

- **Resolve the following sentences using the Resolution Refutation Rule.**

  1- $A \vee B$

  2- $\neg B$

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Method 3: Inference By Resolution Refutation

EXAMPLE

- **Resolve the following sentences using the Resolution Refutation Rule.**

  1- $A \vee B \vee \neg C \vee D$

  2- $\neg A \vee \neg C \vee \neg F$

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Conjunctive Normal Form (CNF)

To use the resolution refutation method, all sentences must be written in (CNF) conjunctive normal form (clause normal form).

A formula is in conjunctive normal form if it is a conjunction of one or more clauses.

EXAMPLE

CNF	Not CNF
• $\neg p$	• $\neg p \rightarrow r$
• $p \vee \neg q$	• $p \leftrightarrow \neg q$
• $r \vee \neg t \vee \neg p$	• $r \wedge (\neg t \vee \neg p)$

Conjunctive Normal Form (CNF)

EXAMPLE

- **Convert the following sentence to CNF.**

  $A \rightarrow B$

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Conjunctive Normal Form (CNF)

EXAMPLE

- **Convert the following sentence to CNF.**

  $A \wedge \sim B$

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Conjunctive Normal Form (CNF)

EXAMPLE

- **Convert the following sentence to CNF.**

  1- $A \leftrightarrow B$

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Conjunctive Normal Form (CNF)

EXAMPLE

- **Convert the following sentence to CNF.**

  $(A \wedge B) \vee C$

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Resolution Refutation Algorithm

1. *Add negation of query to KB*
2. *Convert all sentences in KB and query to CNF.*
3. *Pick 2 sentences that haven't been used before and can be used with the Resolution Rule of inference.*
4. *If none, halt and answer that the query S is **NOT entailed by KB.***
5. *Compute resolvent and add it to KB.*
6. *If False in KB Then halt and answer that the query S is **entailed by KB***

   Else Goto 2

Method 3: Inference By Resolution Refutation

EXAMPLE

- Assume KB has the following sentences:
  1. $B \wedge C \rightarrow A$
  2. B
  3. $D \wedge E \rightarrow C$
  4. $E \vee F$
  5. $D \wedge \neg F$
- Prove that KB |= A using the resolution refutation method.

I will solve this question in two ways

Method 3: Inference By Resolution Refutation

1. Add negation of query to KB

KB	S
1. $B \wedge C \rightarrow A$ 2. B 3. $D \wedge E \rightarrow C$ 4. $E \vee F$ 5. $D \wedge \neg F$	6. A

New KB

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Method 3: Inference By Resolution Refutation

2. Convert all sentences in KB to CNF

KB

KB in CNF

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3. Resolution Rule of Inference

KB in CNF

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Method 3: Inference By Resolution Refutation

Another way of solution

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Method 3: Inference By Resolution Refutation

EXAMPLE

- Assume KB has the following sentences:
  1. $P \rightarrow R$
  2. $Q \rightarrow R$
  3. $P \vee Q$
- Prove that KB $\models$ R using the resolution refutation method.

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Method 3: Inference By Resolution Refutation

1. Add negation of query to KB

KB	S
1. $P \rightarrow R$ 2. $Q \rightarrow R$ 3. $P \vee Q$	4. R

New KB

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Method 3: Inference By Resolution Refutation

2. Convert all sentences in KB to CNF

KB

KB in CNF

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Method 3: Inference By Resolution Refutation

3. Resolution Rule of Inference

KB in CNF

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Method 3: Inference By Resolution Refutation

Another way of solution

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Quiz

- Assume KB has the following sentences:
  1. $(P \rightarrow Q) \rightarrow Q$
  2. $(P \rightarrow P) \rightarrow R$
  3. $(R \rightarrow S) \rightarrow \neg(S \rightarrow Q)$
- Prove that KB $\models$ R using the resolution refutation method.

Quiz

- Assume KB has the following sentences:
  1. heads $\rightarrow$ win(Me)
  2. tails $\rightarrow$ lose(You)
  3. $\neg$heads $\rightarrow$ tails
  4. lose(You) $\rightarrow$ win(Me)
- Prove that KB |= win(me) using the resolution refutation method.

Quiz

- If it is sunny today, then the sun shines on the screen. If the sun shines on the screen, the blinds are brought down. The blinds are not down.

- Prove that it is sunny today using the resolution refutation method.

Check - Chapter Learning Outcomes

CLO

2

- Determine if a given sentence is a proposition or not.
- Complete and use truth tables for expressions involving the following logical connectives: negation, conjunction, disjunction, conditional, and biconditional.
- Write the converse, contrapositive and inverse for a given sentence.
- Translate an English sentence to PL and vice-versa.
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End of Chapter

2